HIWALOY
Software Requirements Specification (SRS)

Project Name: HIWALOY – "How It Will ACTUALLY Look On You"
Project Type: AI-Powered Fashion Fit & Purchase Risk Analysis Platform
Platform: Web Application
Target Domain: E-Commerce / Fashion Technology

HIWALOY is a multimodal AI-powered web platform designed to reduce clothing purchase uncertainty, improve online shopping confidence, and decrease return rates in e-commerce. The platform analyzes user body characteristics, clothing properties, customer reviews, and fit preferences to generate explainable personalized fit recommendations.

1. Purpose

The purpose of HIWALOY is to help users understand how clothing items are likely to fit and appear on their own body before purchasing them online. The system minimizes purchase regret and return risk through AI-driven fit analysis and explainable recommendations.

2. Scope

The system includes: User profile management Body image analysis Clothing image analysis AI-generated fit recommendations Purchase risk estimation Community review retrieval using RAG Recommendation history management

3. Product Perspective

HIWALOY functions as an intelligent assistant integrated into online shopping workflows. The system combines multimodal AI, retrieval-augmented generation (RAG), and agentic AI workflows to provide accurate and explainable fit analysis.

4. Target Users

Online fashion shoppers Users uncertain about sizing Users frequently returning products Consumers interested in personalized recommendations

5. Technology Stack

Frontend: Next.js + TailwindCSS Backend: FastAPI LLM: Gemini API AI Frameworks: LangChain & LangGraph Vector Database: ChromaDB Database: PostgreSQL

6. System Architecture

The system uses a multi-agent architecture: Body Analyzer Agent Garment Analyzer Agent Review Intelligence Agent Fit Recommendation Agent Purchase Risk Agent History Management Agent

7. Functional Requirements

• The system shall allow users to create a personalized profile.

• The system shall accept user body images.

• The system shall analyze body proportions using AI.

• The system shall accept clothing screenshots or product images.

• The system shall analyze garment fit characteristics.

• The system shall generate personalized fit and size recommendations.

• The system shall retrieve review-based insights using RAG.

• The system shall generate purchase risk scores.

• The system shall store previous analyses.

• The system shall provide explainable recommendation reasoning.

8. Non-Functional Requirements

• System response time shall remain under 10 seconds.

• The platform shall support desktop and mobile devices.

• User data shall be securely stored and encrypted.

• The system shall support concurrent users.

• AI outputs shall remain interpretable and explainable.

• The system shall minimize hallucinated recommendations.

• The UI shall remain user-friendly and accessible.

• The system shall maintain high availability.

9. Detailed Use Case Descriptions

UC-01 — Create User Profile

Field	Description
Use Case ID	UC-01
Use Case Name	Create User Profile
Primary Actor	User
Goal	Create personalized fit profile
Trigger	User clicks Get Started
Preconditions	User is on onboarding page
Postconditions	User profile stored
Main Flow	1. User enters height, weight, fit preferences 2. User uploads body image 3. S

Alternative Flows	If image quality is low, system requests another image
Exception Flows	If AI body detection fails, system shows error message

Functional Requirements

• System shall allow body image upload

• System shall collect body-related information

• System shall store user preferences

Non-Functional Requirements

• Profile creation shall complete under 5 seconds

• User data shall be encrypted

• System shall support mobile uploads

UC-02 — Upload Clothing Item

Field	Description
Use Case ID	UC-02
Use Case Name	Upload Clothing Item
Primary Actor	User
Goal	Provide clothing data for analysis
Trigger	User uploads clothing image
Preconditions	User profile exists
Postconditions	Clothing image processed
Main Flow	1. User uploads product image 2. AI analyzes garment 3. System extracts fit m
Alternative Flows	System asks for clearer image if quality is insufficient
Exception Flows	Unsupported file format error

Functional Requirements

• System shall support image uploads

• System shall classify garment category

• System shall extract fit-related metadata

Non-Functional Requirements

• Image processing shall complete under 7 seconds

• System shall support JPG and PNG formats

• Concurrent uploads shall be supported

UC-03 — Analyze Body Type

Field	Description
Use Case ID	UC-03

Use Case Name	Analyze Body Type
Primary Actor	AI System
Goal	Generate body proportion insights
Trigger	Body image upload
Preconditions	Body image exists
Postconditions	Body analysis generated
Main Flow	1. AI processes image 2. System estimates body proportions 3. Fit tendencies
Alternative Flows	Low confidence analysis warning
Exception Flows	Human body cannot be detected

Functional Requirements

• System shall analyze body proportions

• System shall estimate silhouette type

• System shall generate fit insights

Non-Functional Requirements

• AI output shall avoid offensive language

• Confidence score shall be displayed

• Analysis shall remain consistent

UC-04 — Analyze Garment Fit

Field	Description
Use Case ID	UC-04
Use Case Name	Analyze Garment Fit
Primary Actor	AI System
Goal	Understand garment structure and fit
Trigger	Garment upload
Preconditions	Garment image exists
Postconditions	Garment analysis completed
Main Flow	1. AI analyzes garment image 2. System detects fit type 3. Material behavior e
Alternative Flows	Ambiguous garment style warning
Exception Flows	Unreadable image error

Functional Requirements

• System shall identify fit type

• System shall detect oversize/slim/regular fit

• System shall estimate fabric behavior

Non-Functional Requirements

• Analysis latency shall remain under 8 seconds

• Results shall remain explainable

• System shall generate reproducible outputs

UC-05 — Generate Fit Recommendation

Field	Description
Use Case ID	UC-05
Use Case Name	Generate Fit Recommendation
Primary Actor	User
Goal	Receive personalized recommendation
Trigger	User starts fit analysis
Preconditions	Body and garment analyses completed
Postconditions	Recommendation displayed
Main Flow	1. System combines analyses 2. AI predicts fit appearance 3. Recommended
Alternative Flows	Low confidence recommendation shown
Exception Flows	Recommendation generation failure

Functional Requirements

• System shall generate size recommendations

• System shall explain recommendation reasoning

• System shall generate confidence scores

Non-Functional Requirements

• Recommendation generation shall complete under 10 seconds

• Outputs shall remain interpretable

• Hallucinations shall be minimized

UC-06 — Retrieve Community Insights

Field	Description
Use Case ID	UC-06
Use Case Name	Retrieve Community Insights
Primary Actor	AI System
Goal	Retrieve review-based insights
Trigger	Recommendation generation
Preconditions	Product metadata exists
Postconditions	Review insights displayed
Main Flow	1. System retrieves review embeddings 2. RAG pipeline processes comments
Alternative Flows	Limited review availability notification
Exception Flows	Vector retrieval failure

Functional Requirements

• System shall retrieve review embeddings

• System shall summarize customer feedback

• System shall identify complaint patterns

Non-Functional Requirements

• Retrieval latency shall remain under 5 seconds

• Results shall remain relevant

• Duplicate insights shall be avoided

UC-07 — Calculate Purchase Risk

Field	Description
Use Case ID	UC-07
Use Case Name	Calculate Purchase Risk
Primary Actor	AI System
Goal	Estimate dissatisfaction probability
Trigger	Recommendation completion
Preconditions	Recommendation exists
Postconditions	Risk score displayed
Main Flow	1. AI evaluates mismatch factors 2. Risk score calculated 3. Potential issues d
Alternative Flows	Low-confidence risk warning
Exception Flows	Missing analysis data

Functional Requirements

• System shall calculate purchase risk

• System shall classify risk levels

• System shall explain risk factors

Non-Functional Requirements

• Risk evaluation shall remain explainable

• Predictions shall include confidence metrics

• System shall remain responsive

UC-08 — Save Analysis History

Field	Description
Use Case ID	UC-08
Use Case Name	Save Analysis History
Primary Actor	User
Goal	Access previous analyses

Trigger	Analysis completion
Preconditions	User logged in
Postconditions	Analysis stored successfully
Main Flow	1. System stores recommendation 2. User accesses history page 3. Previous
Alternative Flows	Partial save warning
Exception Flows	Database unavailable

Functional Requirements

• System shall save analysis results

• System shall display recommendation history

• System shall allow reopening analyses

Non-Functional Requirements

• History retrieval shall complete under 3 seconds

• Data shall persist reliably

• Storage system shall be scalable

10. Future Enhancements

3D virtual try-on support Cross-brand size intelligence Fashion stylist AI agent Wardrobe memory system Real-time e-commerce integrations

11. Conclusion

HIWALOY combines multimodal AI, retrieval-augmented generation, and agentic workflows to create a personalized online shopping experience. The platform focuses on reducing return rates, improving purchase confidence, and helping users understand how garments will realistically fit their body type before purchase.